



Graph Algorithm and Applications

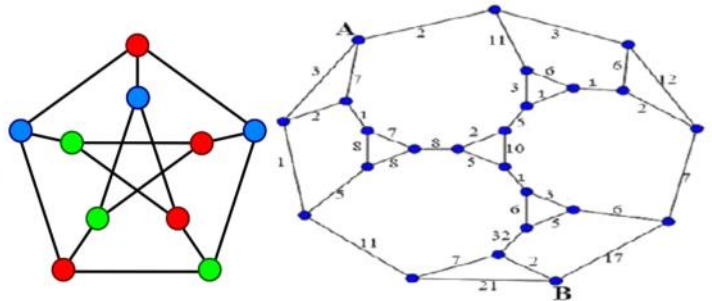
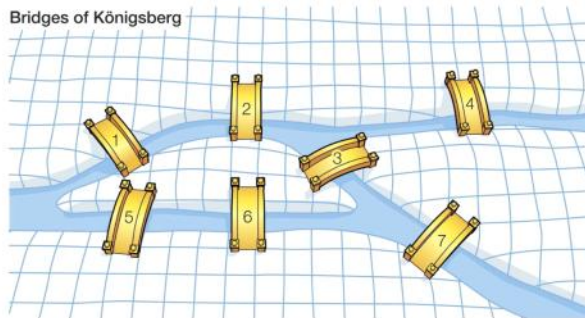
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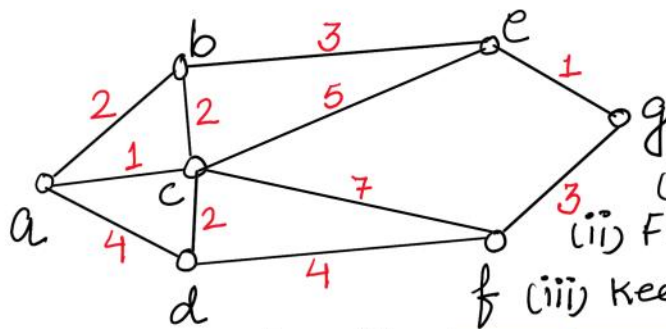
Bridges of Königsberg



Graph Theory: Dijkstra Algorithm

Shortest Path

Dijkstra Algorithm



Find the shortest path from
A $\rightarrow$ g

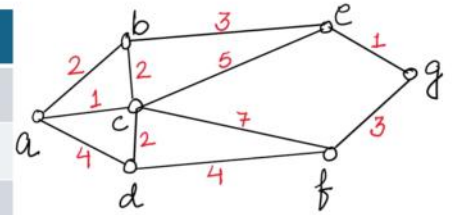
(iv) Add and write the new fixed value to all remaining, if connected.

if the new no. is more then keep the original no.

- (i) Find the distance from 'a'.
(ii) Fix 'a' and then search for next min.
(iii) Keep 0 as such and fix c now.

a	b	c	d	e	f	g
0	2	1	4	∞	∞	∞
0	2	1	3	6	8	∞
0	2	1	3	5	8	∞
0	2	1	3	5	7	∞
0	2	1	3	5	7	6
0	2	1	3	5	7	6

a	b	c	d	e	f	g
0	2	1	4	∞	∞	∞
0	2	1	3	6	8	∞
0	2	1	3	5	8	∞
0	2	1	3	5	7	∞
0	2	1	3	5	7	6
0	2	1	3	5	7	6



Shortest path



a-b-e-g

Shortest distance

from a $\rightarrow$ g

6

APPLICATIONS:

Dijkstra's Algorithm has several real-world use cases, some of which are as follows:

Digital Mapping Services in Google Maps:

Dijkstra's Algorithm is used to find the minimum distance between two locations along the path.

Social Networking Applications:

The standard Dijkstra algorithm can be applied using the shortest path between users measured through handshakes or connections among them.

IP routing to find Open shortest Path First: Open Shortest Path First (OSPF) :

The algorithm provides the shortest cost path from the source router to other routers in the network.

Robotic Path:

Robot/drone moves in the ordered direction by following the shortest path to keep delivering the package in a minimum amount of time.

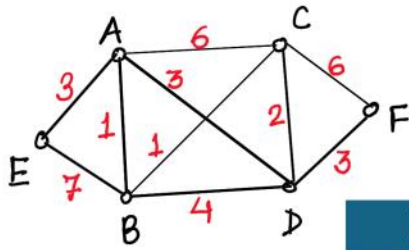
ADVANTAGES & DISADVANTAGES:

Advantages:

1. It is used in Google Maps
2. It is used in finding Shortest Path on geographical Maps.
3. To find locations of Map which refers to vertices of graph. Distance between the location refers to edges.
4. It is used in IP routing to find Open shortest Path First.
5. It is used in the telephone network.

Disadvantages:

1. The major **disadvantage** of the **algorithm** is the fact that it does a blind search there by consuming a lot of time waste of necessary resources
2. It cannot handle negative edges.
3. This leads to acyclic graphs and most often cannot obtain the right shortest path.
4. Time consuming because it expands in every direction



Find the shortest path from $E \rightarrow F$

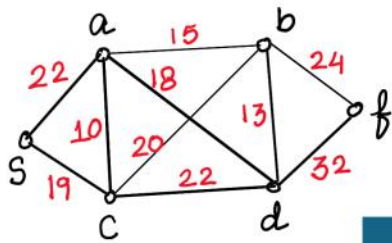
Shortest path

$E-A-D-F$

E	A	B	C	D	F
0	3	7	∞	∞	∞
0	3	4	9	6	∞
0	3	4	5	6	∞
0	3	4	5	6	11
0	3	4	5	6	9

Shortest distance
from E to F

9

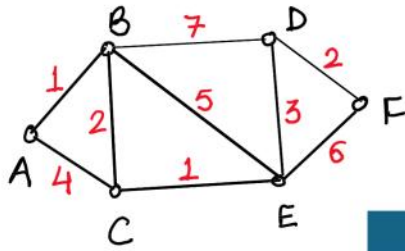


Find the shortest path from $s \rightarrow f$

Shortest path
s-a-b-f

S	A	B	C	D	F
☐	22	∞	19	∞	∞
☐	22	39	19	41	∞
☐	22	37	19	40	∞
☐	22	37	19	40	61
☐	22	37	19	40	61

Shortest distance
from s to F 61

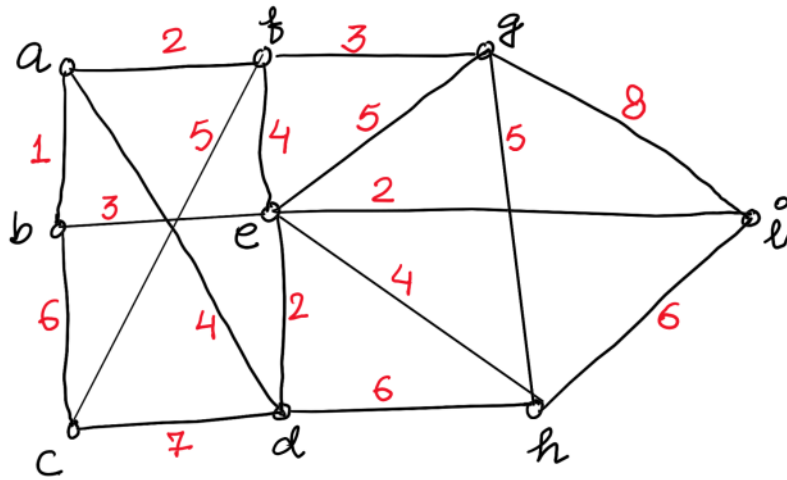


Find the shortest path from $A \rightarrow F$.

Shortest path
a-b-c-e-d-f

A	B	C	D	E	F
0	1	4	∞	∞	∞
0	1	3	8	6	∞
0	1	3	8	4	∞
0	1	3	7	4	10
0	1	3	7	4	9

Shortest distance from A to F $\rightarrow 9$



Find the shortest path from a to i.